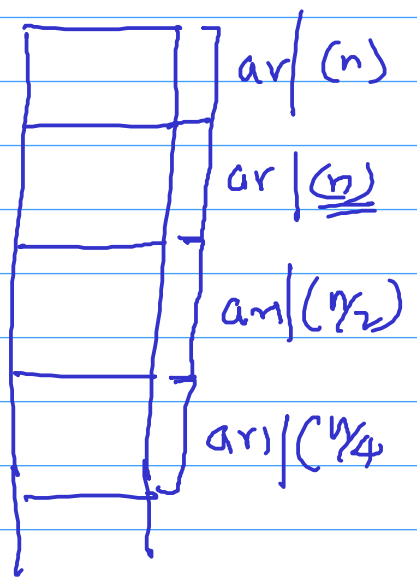
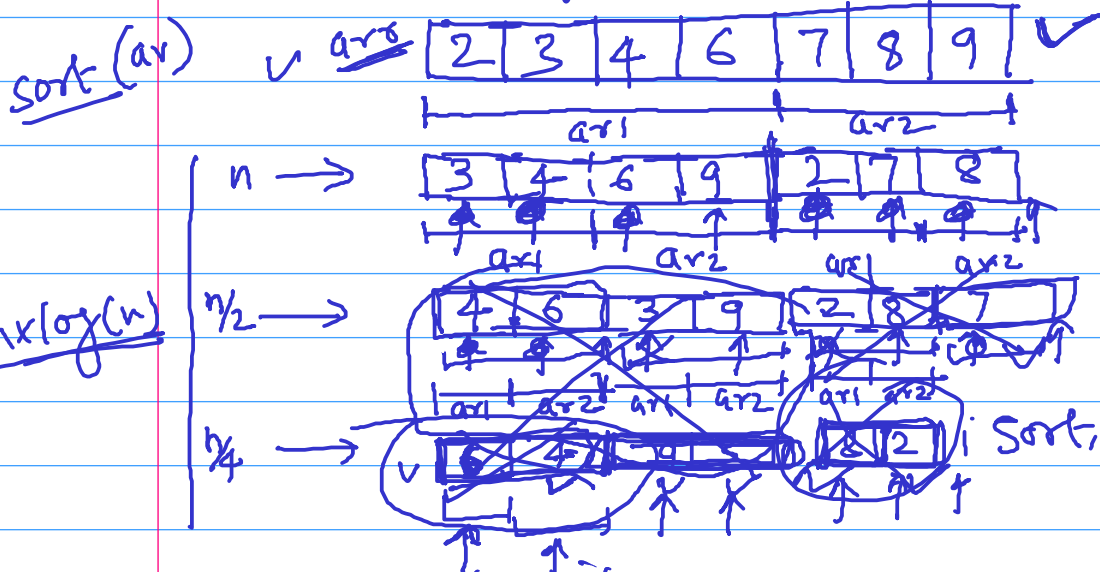


★ Merge sort — Divide and Conquer



Space Complexity

Space required to store partitions

$$n + n/2 + n/4 + \dots + n/n$$

$$n(1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{n})$$

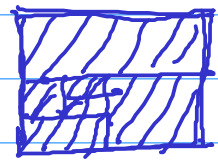
$$\underline{n \times 2} \quad \underline{O(n)}$$

space on stack is $O(\log(n))$
→ Divide ✓

→ Conquer ✓

→ Merge (combine)

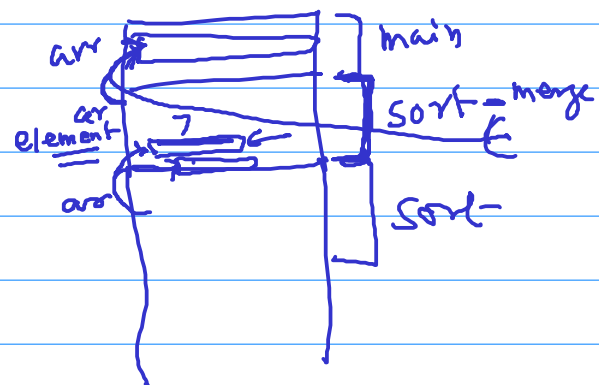
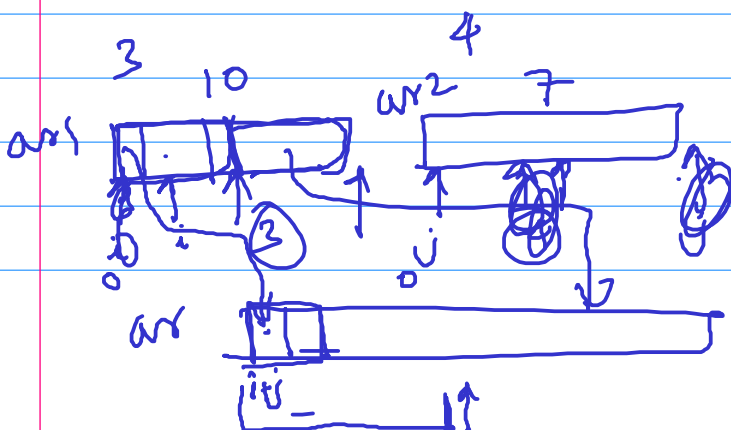
$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{n}$$



Total space = space for storing partitions
+
space on stack

$$= O(n) + O(\log(n))$$

so, $O(n)$ ✓



* Quick sort — Divide and conquer

pivot —

- ① pivot is in final place
- ② $\forall$ left of pivot is smaller or equal to pivot
- ③ $\forall$ right of pivot is greater than pivot

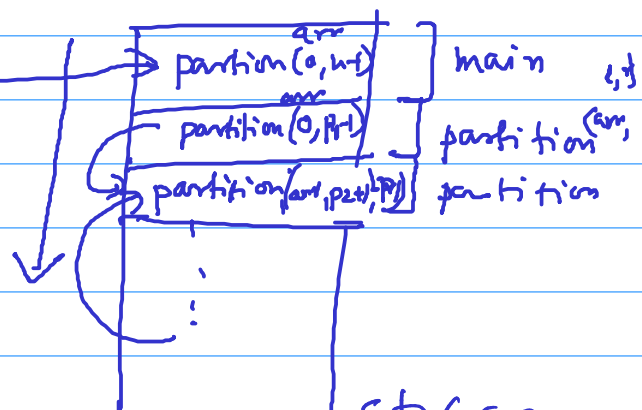
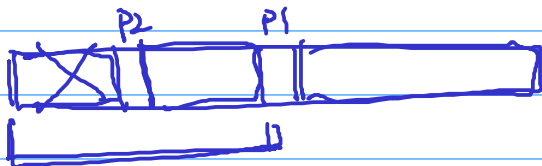
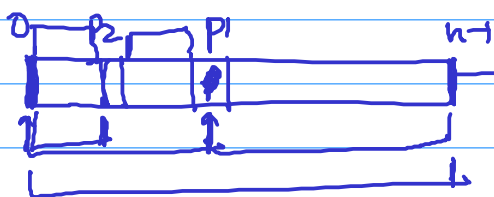
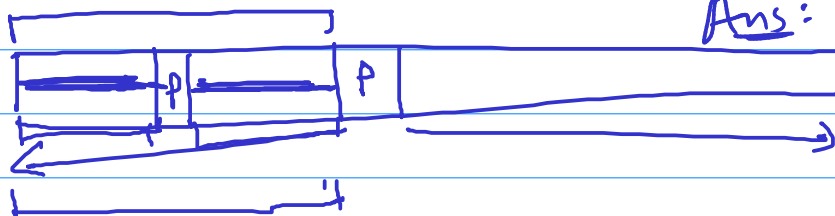
divide — partitioning

conquer —

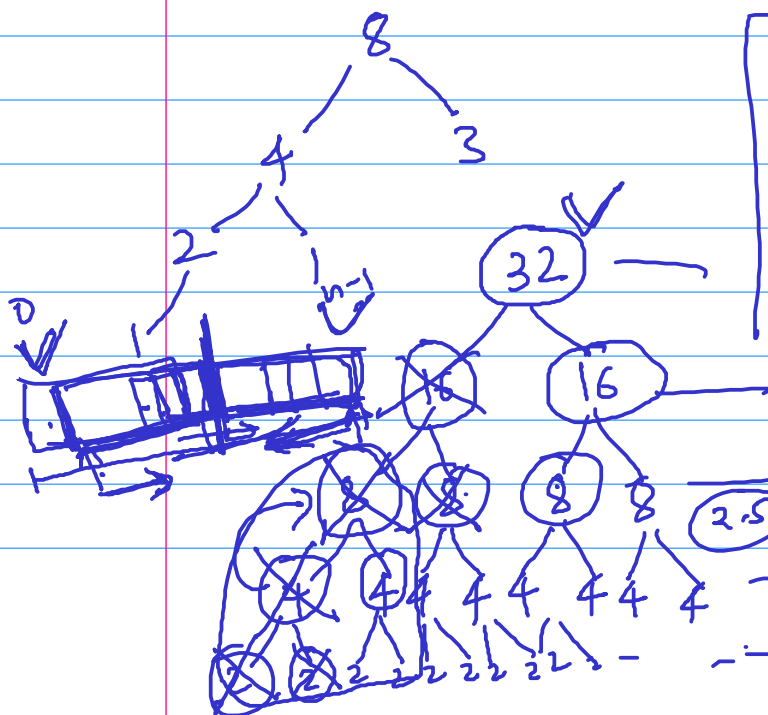
combine —

Que: Do we need to create partition in separate memory like merge sort?

Ans: No. Because partitioning is done on original array



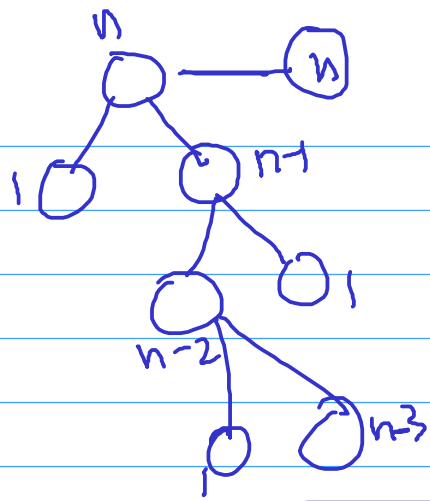
space complexity $O(\log n)$



Partition

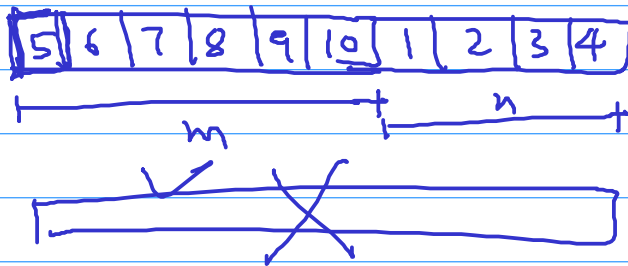
- ① select first element as pivot
- ② Have two pointers starting from left and right
- ③ move left pointer to right until element $>$ pivot
- ④ move right pointer to left until element $<$ pivot
- ⑤ swap

Worst case
time complexity
 $O(n^2)$



$O(n)$
Space complexity in
worst case

Why do we copy array in merge sort?
m+n elements



Best

- Bubble sort
- ✓ Selection sort
- Insertion sort
- ✓ Heap sort
- ✓ Merge sort
- ✓ Quick sort

in-place sorting
stable sorting